Credit Card Fraud Detection Project Report

Internship: InLighnX - Python Development

Project: Credit Card Fraud Detection using Machine Learning

1. Objective

To build a machine learning model capable of identifying fraudulent credit card transactions using historical transaction data.

2. Technologies Used

Python, Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, Joblib, VS Code.

3. Dataset

The project uses the Kaggle Credit Card Fraud Detection dataset containing 284,807 transactions with 31 columns. The target column is 'Class', where 0 represents a normal transaction and 1 represents a fraudulent transaction.

4. Methodology

Loaded the dataset using Pandas. Explored the data and checked for missing values. Visualized class distribution. Separated features and target variable. Split the dataset into training and testing sets. Trained a Random Forest classifier. Generated predictions on the test dataset. Evaluated the model using Accuracy, Classification Report, and Confusion Matrix. Saved the trained model using Joblib.

5. Results

The model successfully classified credit card transactions as either normal or fraudulent. Performance was evaluated using accuracy, precision, recall, F1-score, and a confusion matrix.

6. Project Structure

dataset/ images/ fraud_detection.py credit_card_fraud_model.pkl README.md requirements.txt

7. Conclusion

This project demonstrates the application of machine learning to detect fraudulent financial transactions. Random Forest provided reliable performance on this highly imbalanced dataset.

8. Future Scope

Apply SMOTE or other balancing techniques. Compare multiple machine learning algorithms. Deploy the model using Flask or Streamlit. Integrate real-time fraud detection APIs.